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AI-ENABLED SMART CAMPUS FRAMEWORK FOR PREDICTIVE STUDENT SUPPORT

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Correspondence

Dr. A. Kumar[1],

Dr. B. Priya[2],

Mr. C. Arun[3]

[1]Department of Computer Science, ABC University, India |
[2]Department of Information Technology, XYZ College, India |
[3]Research Scholar, ABC University, India
Corresponding Author: author@example.com

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ABSTRACT

Smart campuses generate large volumes of academic and administrative data, but many institutions still rely on reactive methods to identify students who require academic support. This paper presents a sample AI-enabled smart campus framework that integrates attendance, assessment performance and learning activity indicators to predict students who may require early intervention. The proposed framework uses data preprocessing, feature engineering and a supervised machine-learning classifier. A hypothetical experimental dataset is used only to demonstrate the manuscript structure and reporting style. The sample evaluation reports accuracy, precision, recall and F1-score and illustrates how model results can be compared with baseline approaches. The framework provides a structured workflow for data collection, prediction, intervention and feedback. The study demonstrates how explainable predictive analytics can support evidence-based academic decision-making while maintaining appropriate privacy and ethical safeguards.

Keywords: Artificial Intelligence; Smart Campus; Student Analytics; Predictive Modeling; Machine Learning

1. Introduction

Higher education institutions increasingly collect digital information related to attendance, examinations, assignments and learning management systems. Transforming these records into actionable insights can help institutions identify students who may benefit from timely academic support. Traditional monitoring approaches often depend on manual analysis and may not detect emerging patterns early enough.

This study proposes a conceptual AI-enabled framework for predictive student support. The objective is to demonstrate an end-to-end workflow covering data preparation, prediction, evaluation and intervention. The contribution is a reusable framework structure rather than a claim of validated real-world performance.

2. Related Work / Literature Review

Previous studies have explored machine learning for student performance prediction using attendance, assessment scores, engagement measures and demographic variables. Common approaches include decision trees, random forests, support vector machines and neural networks. However, prediction quality alone is insufficient for institutional deployment; interpretability, privacy, fairness and reproducibility must also be considered.

The research gap addressed in this sample is the integration of prediction with an interpretable intervention workflow. A practical system should not only classify risk but also provide information that academic staff can use to plan appropriate support.

3. Research Gap and Objectives

Research Gap: Existing predictive approaches may focus primarily on model accuracy without presenting a complete and transparent workflow for academic intervention.

Objectives: (1) design an integrated predictive-support framework; (2) define a reproducible preprocessing and evaluation workflow; (3) compare a proposed classifier with baseline methods; and (4) identify ethical and privacy considerations for institutional use.

4. Methodology

For demonstration, a hypothetical dataset containing 2,000 student records is considered. Features include attendance percentage, continuous-assessment average, assignment completion rate and learning-platform activity. Missing values are handled using median imputation for numerical variables, followed by feature scaling where required.

The sample workflow compares Logistic Regression, Decision Tree and Random Forest classifiers. The data are divided into training and testing subsets using an 80:20 split. Performance is reported using accuracy, precision, recall and F1-score. These numerical values are illustrative and must be replaced by results from the actual study before publication.

5. Proposed System / Model

The proposed architecture contains five major stages: data acquisition, preprocessing, predictive modeling, explanation and intervention. Institutional data are transformed into model-ready features. The prediction module estimates support requirements, while the explanation layer identifies influential features. Academic staff can then review the prediction and select an appropriate intervention.

Table 1. Illustrative Model Performance

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	82.5%	80.1%	78.9%	79.5%
Decision Tree	84.0%	82.7%	81.8%	82.2%
Random Forest	88.5%	87.1%	86.4%	86.7%

6. Results and Discussion

In the illustrative experiment, the Random Forest model produces the highest reported F1-score among the three baseline methods. This example demonstrates the expected reporting pattern: numerical results should be accompanied by interpretation, baseline comparison and discussion of limitations.

In an actual research article, the authors should provide the complete dataset description, experimental settings, confidence intervals where appropriate, statistical significance analysis when applicable, and sufficient information for independent reproduction.

7. Conclusion

This sample paper demonstrates how an AI-enabled predictive-support study can be organized using the IJIDCR two-column manuscript structure. The proposed workflow connects data preparation, machine learning, explainability and academic intervention. The reported numerical results are illustrative only and are not intended as evidence of real-world performance.

8. Future Work

Future work may evaluate the framework using real institutional datasets, investigate fairness and model drift, incorporate explainable AI techniques, and conduct controlled studies to measure whether early interventions improve student outcomes.

Data and Code Availability

Sample statement: Data and code availability should be provided by the authors where applicable. For this sample manuscript, no research dataset or code repository is being claimed.

Ethics Statement

Sample statement: Ethical approval and informed consent requirements should be assessed according to the actual study design and institutional policy.

Conflict of Interest

The authors declare that they have no conflict of interest.

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Author Contributions

Author 1: Conceptualization and methodology. Author 2: analysis and review. Author 3: documentation and validation.

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